

PROJECT: ANALYZING MOTORCYCLE PART SALES





You're working for a company that sells motorcycle parts, and they've asked for some help in analyzing their sales data!

They operate three warehouses in the area, selling both retail and wholesale. They offer a variety of parts and accept credit cards, cash, and bank transfer as payment methods. However, each payment type incurs a different fee.

The board of directors wants to gain a better understanding of wholesale revenue by product line, and how this varies month-to-month and across warehouses. You have been tasked with calculating net revenue for each product line and grouping results by month and warehouse. The results should be filtered so that only "Wholesale" orders are included.

They have provided you with access to their database, which contains the following table called `sales` :

Sales

Column	Data type	Description
<code>order_number</code>	<code>VARCHAR</code>	Unique order number.

Column	Data type	Description
<code>date</code>	<code>DATE</code>	Date of the order, from June to August 2021.
<code>warehouse</code>	<code>VARCHAR</code>	The warehouse that the order was made from— <code>North</code> , <code>Central</code> , or <code>West</code> .
<code>client_type</code>	<code>VARCHAR</code>	Whether the order was <code>Retail</code> or <code>Wholesale</code> .
<code>product_line</code>	<code>VARCHAR</code>	Type of product ordered.
<code>quantity</code>	<code>INT</code>	Number of products ordered.
<code>unit_price</code>	<code>FLOAT</code>	Price per product (dollars).
<code>total</code>	<code>FLOAT</code>	Total price of the order (dollars).
<code>payment</code>	<code>VARCHAR</code>	Payment method— <code>Credit card</code> , <code>Transfer</code> , or <code>Cash</code> .
<code>payment_fee</code>	<code>FLOAT</code>	Percentage of <code>total</code> charged as a result of the <code>payment</code> method.

Your query output should be presented in the following format:

<code>product_line</code>	<code>month</code>	<code>warehouse</code>	<code>net_revenue</code>
<code>product_one</code>	---	---	---
<code>product_one</code>	---	---	---
<code>product_one</code>	---	---	---
<code>product_one</code>	---	---	---
<code>product_one</code>	---	---	---
<code>product_one</code>	---	---	---
<code>product_two</code>	---	---	---
...	...	...	...



Projects Data DataFrame as revenue_by_product_line

```

SELECT
  product_line,
  CASE
    WHEN EXTRACT('month' from date) = 6 THEN 'June'
    WHEN EXTRACT('month' from date) = 7 THEN 'July'
    WHEN EXTRACT('month' from date) = 8 THEN 'August'
  END AS month,
  warehouse,
  SUM(total) - SUM(payment_fee) AS net_revenue
FROM sales
WHERE client_type = 'Wholesale'
  AND EXTRACT('month' from date) IN (6, 7, 8)
GROUP BY product_line, month, warehouse
ORDER BY product_line, month, net_revenue DESC;

```

index	...	↑↓	product_line	...	↑↓	mon...	...	↑↓	warehouse
		0	Braking system			August			Central
		1	Braking system			August			West
		2	Braking system			August			North
		3	Braking system			July			Central
		4	Braking system			July			West
		5	Braking system			July			North
		6	Braking system			June			Central
		7	Braking system			June			North
		8	Braking system			June			West
		9	Electrical system			August			North
		10	Electrical system			August			Central
		11	Electrical system			August			West
		12	Electrical system			July			Central
		13	Electrical system			July			North
		14	Electrical system			July			West
		15	Electrical system			June			Central

Rows: 48

[↗ Expand](#)